Experiment No. 1

Title: Implementation of Basic Python Operations and TensorFlow 2 Tensor Operations for Generative AI Applications.

Aim: To understand and implement basic Python operations and TensorFlow 2 tensor operations for building a strong foundation in Generative Artificial Intelligence applications.

Objectives:

1. To understand the basic concepts and functionalities of TensorFlow 2.
2. To learn the concept of tensors and their role in Artificial Intelligence and Generative AI.
3. To create scalar, vector, matrix, and higher-dimensional tensors using TensorFlow.
4. To perform basic tensor operations such as arithmetic calculations, reshaping, and mathematical functions.
5. To understand tensor properties including shape, rank, and data type.
6. To convert TensorFlow tensors into NumPy arrays and compare their functionalities.
7. To develop the programming skills required for implementing advanced Generative AI models.

Pre-Requisites:

1. Knowledge of basic Python programming (variables, data types, loops, functions, and lists).
2. Understanding of basic mathematical operations.
3. Familiarity with Python package installation using pip.
4. Basic understanding of Machine Learning and Artificial Intelligence concepts.
5. Knowledge of NumPy arrays is desirable.

Theory:

TensorFlow is one of the most popular open-source deep learning frameworks developed by Google for building Artificial Intelligence and Machine Learning applications. It provides a comprehensive ecosystem for designing, training, and deploying deep learning models efficiently on CPUs, GPUs, and TPUs.

The fundamental data structure used by TensorFlow is called a Tensor. A tensor is a multidimensional mathematical object that stores numerical data in one or more dimensions. Depending on the number of dimensions, tensors are classified as scalar (0-D), vector (1-D), matrix (2-D), and higher-dimensional tensors (3-D and above).

Unlike ordinary Python lists, TensorFlow tensors are optimized for high-speed numerical computation and parallel processing, making them suitable for large-scale AI applications. Every input processed by modern Generative AI models such as ChatGPT, Gemini, Stable Diffusion, DALL·E, and MusicGPT is internally represented in the form of tensors.

TensorFlow provides numerous built-in operations for creating tensors, performing mathematical computations, reshaping tensors, manipulating dimensions, and converting tensors into NumPy arrays. These capabilities form the mathematical foundation of Deep Learning and Generative AI models.

This experiment introduces students to TensorFlow by demonstrating the creation of tensors, performing tensor operations, understanding tensor properties, and applying basic mathematical functions. The concepts learned in this experiment serve as the basis for implementing advanced Generative AI models such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Transformers, and Large Language Models (LLMs).

Procedure:

1. Import the required libraries such as TensorFlow and NumPy.
2. Verify the TensorFlow installation and display its version.
3. Create scalar, vector, matrix, and higher-dimensional tensors using TensorFlow.
4. Display tensor properties such as shape, rank, and data type.
5. Perform basic tensor arithmetic operations including addition, subtraction, multiplication, and division.
6. Apply TensorFlow mathematical functions such as square, square root, mean, and maximum.
7. Reshape tensors into different dimensions using TensorFlow functions.
8. Convert TensorFlow tensors into NumPy arrays and compare the outputs.
9. Execute the programs and observe the generated outputs.
10. Analyze the importance of tensors in Deep Learning and Generative AI applications.

Conclusion:

The experiment successfully demonstrated the implementation of basic Python operations and TensorFlow 2 tensor operations. Students learned how to create tensors of different dimensions, inspect their properties, perform mathematical operations, reshape tensors, and convert tensors into NumPy arrays. The experiment also highlighted the significance of tensors as the core data structure in TensorFlow and their essential role in building Deep Learning and Generative AI applications. This practical provides the fundamental knowledge required for implementing advanced Generative AI architectures such as GANs, Transformers, and Large Language Models in subsequent laboratory experiments.